ALI SOLTANINEZHAD

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PROFILE

Computer vision PhD candidate at the University of Victoria (expected April 2027) who builds video models that hold up outside the lab: occlusion, glare, motion blur, low frame rates and limited compute. First author of three oral papers (ECCV Workshops 2026, ICPR 2026, CRV 2025), with models running in Archipelago Marine Research's FishVue AI platform. Trained as an electrical engineer (BSc, MSc) in electronics and digital systems.

EDUCATION

Ph.D., Electrical and Computer Engineering, University of Victoria Jan 2023 – Apr 2027 (expected)
Supervisor: Dr. Alexandra Branzan Albu | GPA 9.0 / 9.0 | Advanced to candidacy Nov 2025

- Relevant coursework: Computer Vision (A+), Medical Image Processing (A+), Data Mining (A+)

M.Sc., Electrical Engineering (Digital Electronic Systems), Iran University of Science and Technology Sep 2018 – Jul 2022
Thesis: Human Action Recognition in Still Image Using Transfer Learning | GPA 17.80 / 20

B.Sc., Electrical Engineering (Electronics), Ferdowsi University of Mashhad Sep 2014 – Sep 2018

EXPERIENCE

PhD Researcher, Computer Vision Lab, University of Victoria Jan 2023 – Present
Industry partnership with Archipelago Marine Research (FishVue AI electronic monitoring platform)

- Designed TubeLite (first author, ICPR 2026, oral), a lightweight multi-actor action detector built for temporal stability rather than model size: +4.5 and +7.1 video-mAP@0.5 over the best compared method on MultiSports and UCF101-24, with no optical flow or heavy transformers.
- Kept TubeLite small and fast: 15.65M parameters and 62 GFLOPs per clip, about 4× fewer than the transformer-based TubeR, at 274 frames/s on one H100, while scoring 13.3 points higher video-mAP@0.5 on UCF101-24.
- Led TrapCounter (first author, CRV 2025, oral), a detection, tracking and optical-flow pipeline that counts active fishing traps in deck-camera video, cutting counting error from 16.2 to 0.8 traps per video against manual counts; deployed on commercial fishing vessels.
- Created and released DTCFT, the first public dataset for detecting, tracking and counting fishing traps in vessel footage: 10 videos, 16,500 frames with tracking IDs and 3,300 detection keyframes.
- Developed error-aware reviewer guidance (first author, ECCV 2026 Marine Vision workshop, oral) that aims a fixed 10% human-review budget at the predictions most likely to be wrong: the audited record kept 55 of 57 events, against 50 for the detector alone and 52 with random review, with no retraining.
- Delivered models that run in Archipelago's FishVue AI platform and built the pipeline from raw vessel footage to live predictions; FishVue AI won the 2026 VIATEC Innovative Excellence Award.
- Built and delivered to Archipelago an anonymization pipeline that detects, tracks and masks people across long vessel footage; paper in preparation.
- Built a real-time, motion-aware detector for rare discard events in long commercial fishing videos, designed for at-sea hardware, heavy class imbalance and occlusion.
- Built a query-by-example video search tool on V-JEPA self-supervised embeddings and clustering, and am extending it toward open-set detection of unknown actions.

Applied AI Specialist (part-time, remote), Wireless PnC Inc. Sep 2026 – Present
Wireless charging for robotics and electric mobility

- Developing AI-assisted foreign and living object detection and alignment guidance for wireless charging pads, combining electromagnetic parameter estimation with computer vision.
- Developing fault detection and predictive maintenance from charger telemetry, using anomaly detection to catch component degradation early.

Research Assistant, Computer Vision Lab, Iran University of Science and Technology 2018 – 2022

- Designed and trained the deep models in PyTorch for a multi-expert human action recognition framework with hierarchical super-class learning (Knowledge-Based Systems, 2022).
- First-authored a transfer learning study of action recognition in still images (ICCKE 2021), the basis of my MSc thesis.
- Created and released IHARD46, a 46-class imbalanced human action recognition dataset.

TEACHING

Sessional Instructor, ECE 435 / BME 403 Medical Image Processing, University of Victoria Jan – Apr 2026

- Sole instructor for a 15-student cross-listed 4th-year and graduate course; created all lectures, assignments, the final project and exams.
- Restructured the course to teach problem formulation and spatial and temporal reasoning before deep learning, supported by the President's Fellowship in Research-Enriched Teaching.

Teaching Assistant and Lab Instructor, Electrical and Computer Engineering, University of Victoria May 2023 – Present

- Super TA for ECE 471 Computer Vision (2024) and ECE 537 Applied Data Analysis (4 terms, 2025 to 2026).
- TA and lab instructor for 13 offerings of 9 ECE courses, including Digital Design, VLSI Systems, Electronic Circuits, Control Systems, and Signals and Systems: ran labs and tutorials, graded, held office hours and mentored students on course projects.

PUBLICATIONS

- A. Soltaninezhad, M. Cote, A. Rico Espinosa, T. Porto Marques, A. Branzan Albu, J. Lower. Error-Aware Reviewer Guidance for AI-Assisted Fisheries Electronic Monitoring. 2nd Workshop on Marine Vision, ECCV Workshops, 2026. Oral.
- A. Soltaninezhad, M. Cote, A. Rico Espinosa, T. Porto Marques, A. Branzan Albu. TubeLite: Lightweight Multi-Actor Spatio-Temporal Action Detection. International Conference on Pattern Recognition (ICPR), 2026. Oral.
- A. Soltaninezhad, M. Cote, A. Rico Espinosa, T. Porto Marques, A. Branzan Albu, J. Paetkau, V. Diaz, J. Lower. TrapCounter: Integrating Detection, Tracking and Optical Flow for the Automatic Counting of Active Fishing Traps. Conference on Robots and Vision (CRV), 2025. Oral.
- A. Rico Espinosa, M. Cote, A. Soltaninezhad, T. Porto Marques, A. Branzan Albu, V. Diaz Gimeno, J. W. Lower, R. Prussin. Quilting-based Image Pre-processing for Commercial Ground Hook and Line Fishing Imagery Classification. ICPR, 2026.
- H. A. Dehkordi, A. Soltaninezhad, H. Kashiyani, S. B. Shokouhi, A. Ayatollahi. Multi-Expert Human Action Recognition with Hierarchical Super-Class Learning. Knowledge-Based Systems, 2022.
- A. Soltaninezhad, H. A. Dehkordi, S. S. Ashrafi, S. B. Shokouhi. To Transfer or Not To Transfer (TNT): Action Recognition in Still Image Using Transfer Learning. ICCKE, 2021.
- H. A. Dehkordi, A. Soltaninezhad, S. S. Ashrafi, S. B. Shokouhi. Still Image Action Recognition Using Ensemble Learning. International Conference on Web Research (ICWR), 2021.
- H. A. Dehkordi, A. Soltaninezhad, S. B. Shokouhi. Robust Road Segmentation of Self-Driving Cars Using Deep Learning. Conference on Recent Developments and Future Trends in the Automotive Industry, Tehran, 2021.

SKILLS

Programming: Python, MATLAB, Git, LaTeX

Machine learning: PyTorch, TensorFlow / Keras, scikit-learn, OpenCV, NumPy, Pandas, Hugging Face

Computer vision: spatio-temporal action detection, multi-object tracking, optical flow, self-supervised video embeddings (V-JEPA), vision-language models, transfer learning

Compute and tooling: multi-GPU training, Slurm on Digital Research Alliance of Canada clusters, Weights & Biases, MLflow, TensorBoard, Docker, GitHub Actions

Engineering foundations: analog and digital electronics, digital system design, signals and systems, control systems

Languages: English (full professional), Persian (native)

HONOURS, AWARDS AND SERVICE

- President's Fellowship in Research-Enriched Teaching, University of Victoria ($10,000), 2025
- UVic nominee, 2026 Google PhD Fellowship
- Participant, AI in Finance Summer School 2026, hosted by RBC Borealis, Vancouver
- Top 1% (BSc) and top 3% (MSc), Iran national university entrance exams
- Reviewer, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)